CHAPTER - 1

INTRODUCTION

Dyslexia and Attention-Deficit/Hyperactivity Disorder (ADHD) are two common neurodevelopmental conditions that significantly affect a child's learning ability, behavior, and cognitive development. Dyslexia primarily impacts reading, spelling, and language processing skills, whereas ADHD affects attention, impulse control, and executive functioning. These conditions often coexist, making early identification and intervention even more critical for improving academic performance and overall well-being.

Traditional methods for identifying Dyslexia and ADHD rely heavily on clinical evaluations, observational techniques, and paper-based assessments conducted by specialists. While these methods can provide accurate diagnoses, they are often time-consuming, expensive, and not easily accessible to all individuals, especially in early childhood stages. Moreover, these approaches lack interactivity and may not effectively engage young children, particularly toddlers who cannot read or respond to structured questionnaires.

To address these challenges, the proposed system introduces a web-based, gamified screening platform designed to detect early signs of Dyslexia and ADHD through interactive Continuous Performance Tasks (CPT). The system transforms traditional assessments into engaging, game-like activities that are specifically tailored for different age groups, including toddlers (1–5 years), children, and teenagers. By integrating technology-driven evaluation methods, the platform ensures a more accurate, objective, and user-friendly screening process.

A key innovation of this system is the 3x3 Age-Stratified Game Evaluation Matrix, which includes three core assessment modules: reading, attention, and behavior. The reading test utilizes the Web Speech API to assist non-reading children by converting phonemes into audible instructions, allowing them to interact through image selection. The attention test is developed using an HTML5 Canvas-based game engine, where children respond to moving visual stimuli, enabling the system to measure reaction time and impulsive behavior through metrics such as commission errors. The behavior test incorporates memory-based gameplay, such as object tracking and recall tasks, to evaluate cognitive processing and working memory.

In addition to assessment, the system enhances user experience through neurodivergent friendly UI/UX design improvements. Features such as a persistent dark mode (Focus Mode) reduce visual overstimulation for ADHD users, while a dopamine-based reward system using confetti

animations and audio feedback reinforces positive engagement. These elements ensure that the platform is not only functional but also psychologically supportive for children with attention and learning challenges.

Furthermore, the system includes a comprehensive analytical dashboard powered by Chart.js, which provides real-time visualization of performance metrics and risk levels categorized as Low, Medium, or High. Teachers and parents can access detailed reports and track a child's progress over time, enabling informed decision-making and early intervention.

An additional advancement is the Interactive Virtual Training and Remediation Library, which offers practical tools such as digital focus games, text-to-speech readers, speech-to-text dictation, and Pomodoro timers. These features extend the system beyond detection by actively supporting skill development and behavioral improvement.

Overall, the proposed system represents a significant shift from traditional diagnostic approaches to a modern, scalable, and engaging digital solution. By combining gamification, real-time analytics, and assistive technologies, the platform provides an effective and accessible method for early screening and continuous support for children with Dyslexia and ADHD.

1.1. PROJECT OBJECTIVE

The primary objective of the proposed system is to develop a web-based, gamified screening platform for the early identification of Dyslexia and Attention-Deficit/Hyperactivity Disorder (ADHD) in children. The system aims to replace traditional, non-interactive assessment methods with engaging Continuous Performance Tasks (CPT) implemented through interactive games. By utilizing modern web technologies, the platform enables children to participate in assessments in a natural and enjoyable manner, improving accuracy and participation levels.

The system is designed to assist parents and teachers in identifying early signs of learning and attention disorders by analyzing key performance indicators such as response time, accuracy, and behavioral patterns. It provides automated risk classification into Low, Medium, and High levels, allowing users to understand the child's cognitive condition without requiring immediate clinical intervention. In addition to detection, the system also focuses on improvement by integrating a virtual training and remediation library, offering tools that support skill development in reading, attention, and memory.

The project further aims to ensure accessibility, scalability, and usability by developing a browser-based platform that can be used in both home and school environments. By combining

gamification, real-time analytics, and assistive technologies, the system provides a comprehensive solution for early screening and continuous support.

Objectives of the System:

1. To develop a web-based platform for early screening of Dyslexia and ADHD.
2. To implement gamified Continuous Performance Tasks (CPT) for assessment.
3. To evaluate reading, attention, and behavioral patterns using interactive tests.
4. To analyze performance data such as accuracy, response time, and error patterns.
5. To generate automated risk levels (Low, Medium, High) using algorithmic logic.
6. To provide real-time analytical dashboards using graphical representations.
7. To integrate a virtual training module for improving cognitive skills.
8. To create an engaging, user-friendly, and accessible system for children and educators.

1.2. SCOPE OF THE PROJECT

The scope of the proposed system includes the design and development of a web-based application that facilitates early screening and support for children with potential Dyslexia and ADHD. The system is intended to be used by administrators, teachers, and parents, providing role-based access to different functionalities such as user management, test conduction, and report analysis.

The platform supports children across multiple age groups, particularly focusing on early developmental stages (ages 1–5) as well as older children and teenagers. It includes gamified assessment modules that evaluate reading ability, attention span, and behavioral patterns using interactive techniques. The system captures real-time data during gameplay, processes it using predefined evaluation logic, and generates meaningful insights through analytical dashboards.

In addition to detection, the system extends its scope to include a training and remediation module, which provides interactive tools such as attention games, text-to-speech readers, speech-to-text input, and time management utilities. These features help children improve their cognitive and learning skills over time.

The system is designed to operate on standard web browsers, making it easily accessible without requiring specialized hardware or software. It is suitable for use in educational institutions, homes, and preliminary screening environments.

Scope Includes:

- Web-based platform accessible via browsers

- Role-based system (Admin, Teacher, Parent)
- Gamified assessment modules (Reading, Attention, Behavior)
- Real-time data collection and analysis
- Automated risk classification and reporting
- Graphical dashboards for performance visualization
- Interactive training and remediation tools

Future Scope:

- Development of mobile application (Android/iOS)
- Integration of Machine Learning for predictive analysis
- Advanced behavioral tracking using AI techniques
- Cloud-based data storage and scalability

1.3. PROBLEM STATEMENT & CLINICAL CONTEXT

Dyslexia and ADHD are often underdiagnosed in early stages due to lack of accessible and engaging screening methods. Traditional systems depend on manual observation and clinical testing, which may not always be available or affordable. These methods are also less effective for young children who may not respond well to structured or text-based assessments.

Additionally, existing systems lack interactivity and real-time analysis, making it difficult to accurately measure attention, behavior, and cognitive performance. The absence of engaging tools reduces participation and may lead to incomplete or inaccurate assessments.

Therefore, there is a need for a system that is interactive, accessible, and capable of providing accurate early screening through automated and engaging methods.

1.4. MODULE DESCRIPTION

The system consists of four main modules. The Secure Access Module manages user authentication and role-based access. The Game Engine Module conducts interactive tests to evaluate cognitive skills. The Dashboard Module displays results and performance through charts. The Training Module provides tools to improve cognitive abilities.

- The **Secure Access Module** manages user authentication and authorization for Admin, Teacher, and Parent roles. It ensures secure login, protects user data, and provides role-based access to different functionalities of the system.

- The **3x3 Age-Stratified Game Engine Module** is the core component of the system. It provides interactive, gamified assessments tailored for different age groups. It includes reading, attention, and behavior tests that collect real-time cognitive data such as accuracy, response time, and error patterns.
- The **Chart.js Analytical Dashboard Module** processes the collected data and presents it in the form of graphical reports. It displays risk levels and performance metrics using charts, enabling parents and teachers to easily understand and track the child's progress over time.
- The **Virtual Training and Remediation Library Module** extends the system beyond screening by providing tools for cognitive improvement. It includes features such as speech-to-text, audio reading, attention training activities, and time management tools, helping children develop their skills in an interactive way.

CHAPTER - 2

SYSTEM SPECIFICATION

The system specification defines the technical requirements needed to develop and operate the proposed Dyslexia and ADHD screening system effectively. It includes details about the hardware configuration, software requirements, and technologies used in building the system. These specifications ensure that the application runs smoothly, provides reliable performance, and supports all functionalities such as gamified assessments, real-time analytics, and user management.

2.1. HARDWARE CONFIGURATION REQUIREMENTS

The proposed system is a web-based application and does not require high-end hardware. However, a standard computing system is necessary to ensure smooth execution and performance of the application.

Minimum Requirements:

- **Processor:** Intel i3 or equivalent
- **RAM:** 4 GB (8 GB recommended for better performance)
- **Storage:** 256 GB HDD/SSD
- **Display:** Standard monitor with good resolution
- **Input Devices:** Keyboard, Mouse / Touchscreen
- **Internet Connection:** Required for web access

Recommended Requirements:

- **Processor:** Intel i5 or above
- **RAM:** 8 GB or higher
- **Storage:** SSD for faster data processing
- **Internet:** High-speed broadband connection

2.2. SOFTWARE CONFIGURATION REQUIREMENTS

Software specification includes the tools, frameworks, programming languages, and libraries used to develop and run the system. The application uses a lightweight and efficient technology stack to ensure fast performance and easy deployment.

Frontend Technologies

The frontend is responsible for user interaction and visual presentation of the system. It is designed to be interactive, responsive, and user-friendly, especially for children.

- **HTML5:**
  Used to structure web pages and create the overall layout of the application.
- **CSS3:**
  Used for styling and designing the user interface, including layouts, colors, and responsiveness.
- **JavaScript:**
  Provides dynamic functionality and handles user interactions, game logic, and data processing on the client side.
- **HTML5 Canvas:**
  Used to build interactive game environments such as the attention test where objects move dynamically on the screen.
- **Web Speech API:**
  Enables speech synthesis and recognition. It is used in reading tests to provide audio instructions for non-reading children.
- **Chart.js:**
  A JavaScript library used to generate graphical representations such as pie charts and line charts for performance analysis.
- **Canvas-Confetti Library:**
  Used to provide reward animations (confetti effects) to enhance user engagement and motivation.

Backend Technologies

The backend handles server-side processing, data management, and application logic.

- **Python:**
  The primary programming language used for implementing backend logic due to its simplicity and flexibility.
- **Flask Framework:**
  A lightweight web framework used to build the server-side application. It manages routing, request handling, and communication between frontend and backend.
- **Werkzeug Security (Cryptography):**
  Used for secure password handling and authentication mechanisms.

1) **Database Technologies**

The system uses a relational database to store user information, test results, and performance data.

- **SQLite / MySQL:**
  Used for storing structured data such as user details, test results, and reports.
- **SQL Alchemy ORM:**
  Used to interact with the database using object-oriented programming, simplifying database operations and improving code readability.

2.3. SOFTWARE DESCRIPTION

The software components used in the system work together to provide a seamless and efficient platform for screening and analysis. The backend is developed using Python and Flask, which handle data processing, user authentication, and communication with the database. Flask provides a lightweight and flexible environment for building web applications and managing server-side operations.

The frontend is developed using HTML, CSS, and JavaScript, which create an interactive and visually appealing user interface. Advanced frontend features such as HTML5 Canvas enable the implementation of dynamic game-based assessments, while the Web Speech API allows audio interaction for children who cannot read. These technologies make the system highly engaging and accessible.

The database plays a crucial role in storing user data, test results, and performance metrics. SQL Alchemy ORM is used to simplify database operations and ensure data consistency. The system maintains relationships between different entities such as users, children, and test results, allowing efficient retrieval and analysis of data.

Visualization tools such as Chart.js are used to display analytical results in graphical formats, making it easier for users to interpret performance data. Additionally, the integration of lightweight libraries such as canvas-confetti enhances user experience by providing reward-based feedback mechanisms.

Unlike traditional systems that rely on heavy machine learning frameworks, this system is designed using a lightweight and efficient architecture, ensuring faster performance, easier deployment, and better scalability. This makes the application suitable for use in both educational institutions and home environments.

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CHAPTER - 3

SYSTEM STUDY

The system study is the process of analyzing the existing system and identifying its limitations in order to design an improved and efficient solution. It helps in understanding the challenges faced in traditional methods and provides a foundation for developing a better system. In the context of this project, the system study focuses on evaluating the current approaches used for identifying Dyslexia and Attention-Deficit/Hyperactivity Disorder (ADHD) in children and proposing an enhanced, technology-driven alternative.

3.1. EXISTING SYSTEM

In the existing system, identification of learning disorders such as Dyslexia and ADHD is primarily carried out through traditional assessment methods. These methods include teacher observations, manual reading tests, behavioral questionnaires, and clinical evaluations conducted by psychologists or medical professionals. While these approaches can provide valuable insights, they often rely heavily on subjective judgment and lack standardization.

Teachers typically observe a child's reading ability, attention span, and classroom behavior to identify potential learning difficulties. However, this method depends on individual experience and perception, which may vary from one teacher to another. As a result, early signs of Dyslexia or ADHD may be overlooked or misinterpreted.

Another commonly used approach is paper-based screening tests and questionnaires. These tests require manual evaluation and can be time-consuming, especially when assessing a large number of students. Additionally, these methods are not interactive and may fail to engage children effectively, leading to inaccurate or incomplete assessments.

Clinical assessments conducted by specialists are considered more reliable, but they are often expensive and not easily accessible to all families or schools. These evaluations also require significant time, delaying early detection and intervention.

Drawbacks of the Existing System

- Heavy reliance on manual observation and subjective judgment
- Lack of interactive and engaging assessment methods
- Time-consuming and inefficient evaluation process

- Expensive clinical testing procedures
- Limited accessibility for early-stage detection
- Inconsistent results due to human bias

2) Limitations of Traditional Observational Screening

Traditional screening methods depend on observing behavioral patterns over time, which may not always accurately reflect a child's cognitive abilities. Children may behave differently in different environments, making it difficult to draw consistent conclusions. Moreover, observational methods lack real-time data collection and analytical capabilities, reducing the accuracy and reliability of results.

3.2. PROPOSED SYSTEM

The proposed system is a web-based, gamified screening platform designed to overcome the limitations of traditional methods. It introduces an innovative approach by integrating Continuous Performance Tasks (CPT) into interactive games, making the assessment process both engaging and effective.

The system allows users such as parents and teachers to register, create child profiles, and conduct screening tests through a browser-based interface. The assessments are designed as interactive games that evaluate reading ability, attention span, and behavioral patterns in real-time.

Unlike traditional systems, this platform automatically records performance data such as response time, accuracy, and error patterns. The collected data is processed using predefined evaluation logic to generate risk levels and provide meaningful insights.

Algorithmic Risk Mapping

The system uses an algorithm-based approach to evaluate the child's performance. It analyses key parameters such as:

- **Commission Errors:** Incorrect actions (e.g., clicking wrong objects)
- **Omission Errors:** Missed targets or delayed responses
- **Response Time:** Speed of reaction to stimuli
- **Accuracy:** Correct responses during tasks

Based on these parameters, the system calculates risk levels and categorizes them into:

- **Low Risk**
- **Medium Risk**
- **High Risk**

This automated evaluation reduces human bias and improves accuracy.

Continuous Performance Tasks (CPT) Web Integration

The system implements Continuous Performance Tasks in the form of interactive games. These tasks simulate real-world scenarios where attention and cognitive skills are tested under dynamic conditions.

- The attention test measures focus and impulsivity using moving objects
- The reading test evaluates phonological processing using audio-based interaction
- The behavior test assesses memory and cognitive tracking through gameplay

This integration makes the assessment process more natural and suitable for children, especially those who cannot respond to traditional testing methods.

Advantages of the Proposed System

- Provides an interactive and engaging environment for children
- Reduces dependency on manual observation
- Enables real-time data collection and analysis
- Offers automated and consistent evaluation results
- Accessible through web browsers without specialized tools
- Supports early detection and intervention
- Includes training and remediation features for improvement

CHAPTER - 4

SYSTEM DESIGN

System design defines how the system is structured and how different components interact. The MindTrack system is designed as a web-based application using a layered architecture consisting of frontend, backend, and database.

The frontend handles user interaction and displays the interface, the backend processes logic using Flask, and the database stores user and test data. The system follows the MVC pattern for better organization and scalability.

4.1. FILE DESIGN & COMPONENT ARCHITECTURE (MVC PATTERN)

The system follows the **Model-View-Controller (MVC) architecture**, which separates the application into three main components:

- **Model:**
  Represents the data layer. It includes database tables such as Admin, Teacher, Parent, Child, and Student_Test. It manages data storage and retrieval using SQL Alchemy ORM.
- **View:**
  Represents the user interface. It is developed using HTML, CSS, JavaScript, and Bootstrap. It includes pages such as login, dashboard, test interface, and reports.
- **Controller:**
  Represents the application logic. It is implemented using Flask. It handles user requests, processes data, performs risk calculations, and sends responses to the frontend.

This architecture improves code organization, reduces complexity, and makes the system easier to maintain and extend.

4.2. INPUT DESIGN

The system accepts different types of inputs from users and interactive modules.

Login & Form Inputs

- User credentials (username, password)

- Registration details
- Child information (name, age, gender)

Game-Based Inputs

- Mouse clicks / touch inputs
- Reaction time
- Correct and incorrect responses

Voice Inputs

- Audio interaction using Web Speech API

4.3. OUTPUT DESIGN

The system provides outputs in analytical and interactive formats.

Dashboard Output

- Risk levels (Low / Medium / High)
- Performance scores
- Charts using Chart.js

Gamified Output

- Confetti rewards
- Audio feedback
- Instant results

4.4. DATABASE SCHEMA DESIGN

The system uses a relational database to store and manage data securely. The database maintains relationships between users, children, and test results.

Entity-Relationship (ER) Diagram

The ER diagram represents the relationship between different entities:

- Admin manages Teacher accounts
- Teacher manages Child records
- Parent is linked to Child
- Child is linked to Student_Test records

These relationships ensure proper data organization and integrity.

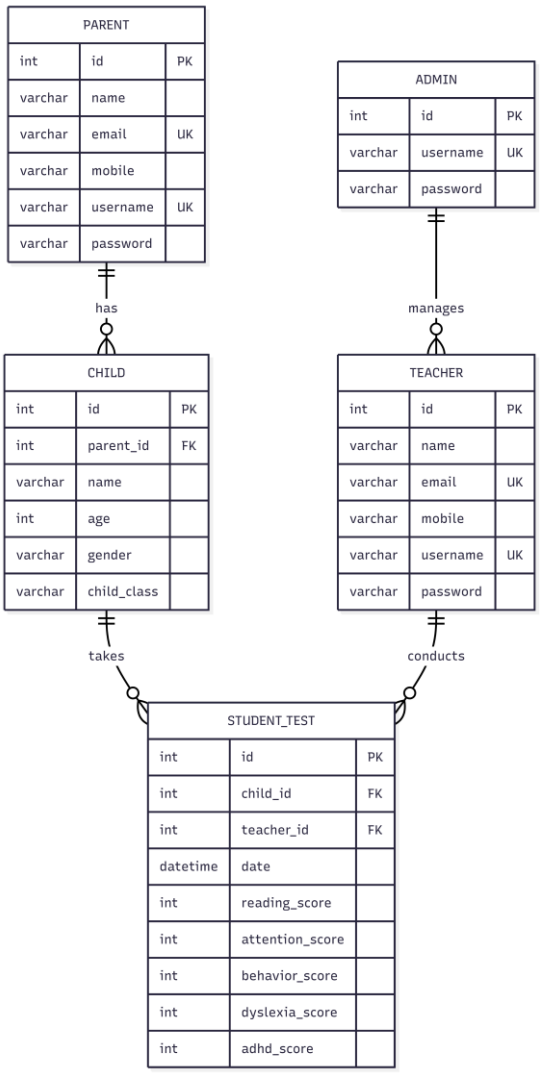


Figure 4.1: Entity Relationship Diagram

Table Structures (Admin, Teacher, Parent, Child, Student_Test)

The database consists of the following main tables:

- **Admin Table:** Stores admin login details
- **Teacher Table:** Stores teacher information
- **Parent Table:** Stores parent details
- **Child Table:** Stores child profiles
- **Student_Test Table:** Stores test results including scores and risk levels.

Each table is connected using primary and foreign keys to maintain consistency.

NOTATIONS

SYMBOL	NAME	DESCRIPTION
	Process block	The Process block represents a step where an action or task occurs, such as a calculation or user interaction. It shows where the system performs a specific operation to progress the flow.
	User Authentication block	The User Authentication block is a Process that represents the functional step where the system validates a user's identity. It performs a specific operation— comparing the provided credentials against the User Database—to determine whether the user can proceed to system dashboard or not.
	Flow line	The Flow Line indicates the direction of the process, showing the sequence in which actions or decisions occur. Arrows connect different steps and guide the flow from one block to another

4.5. DATABASE SCHEMA TABLES

Table 1: student_test (Transactional table storing Game Module Scores)

Field Name	Data Type	Key Type	Description
id	Integer	Primary Key	Unique test log identifier
child_id	Integer	Foreign Key	References id in child table
date	Date Time	-	Timestamp of test completion
reading_score	Integer	-	1-3 metric from Reading tasks
attention_score	Integer	-	1-3 metric from Attention tasks
behavior_score	Integer	-	1-3 metric from Behavior tasks
dyslexia_score	Integer	-	Computed Dyslexia Risk Level
adhd_score	Integer	-	Computed ADHD Risk Level

Table 2: admin (Stores System Administrator credentials)

Field Name	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for the admin
username	Varchar(150)	Unique	Login username for the admin
password	Varchar(150)	-	Hashed password string

Table 3: teacher (Stores Teacher profiles authorized by Admin)

Field Name	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for the Teacher
name	Varchar(150)	-	Full name of the Teacher
email	Varchar(150)	Unique	Contact email
mobile	Varchar(20)	-	Phone contact
username	Varchar(150)	Unique	Login username
password	Varchar(500)	-	Encrypted (bcrypt) hashed password

Table 4: parent (Stores Parent profiles who register dynamically)

Field Name	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for the Parent
name	Varchar(150)	-	Full name of the Parent
email	Varchar(150)	Unique	Contact email
mobile	Varchar(20)	-	Phone contact
username	Varchar(150)	Unique	Login username
password	Varchar(500)	-	Encrypted (bcrypt) hashed password

Table 5: child (Stores Patient/Student demographics linked to a Parent)

Field Name	Data Type	Key Type	Description
id	Integer	Primary Key	Unique identifier for the Child
parent_id	Integer	Foreign Key	References id in parent table
name	Varchar(150)	-	Full name of the Child
age	Integer	-	Determines age group test logic
gender	Varchar(50)	-	Gender of the Child
child_class	Varchar(50)	-	Current educational grade level

CHAPTER - 5

SYSTEM TESTING

System testing is an essential phase in software development that ensures the complete system functions correctly and meets the specified requirements. It involves verifying the integration of all modules and identifying errors before deployment. In the proposed MindTrack: Dyslexia & ADHD Evaluator, testing plays a crucial role in ensuring that all components such as authentication, game-based assessments, data processing, and report generation operate accurately and efficiently. The system was tested under different conditions to validate functionality, performance, usability, and security. Special attention was given to testing interactive modules such as game-based tasks and speech-based inputs, as they directly affect user experience and accuracy of results.

5.1. TESTING METHODOLOGIES

The testing process followed a structured approach to ensure system reliability. Multiple testing techniques were applied, including unit testing, integration testing, functional testing, security testing, and user acceptance testing.

Each module was first tested independently and then tested as part of the complete system. Both manual testing and real-time execution testing were performed to verify system behavior under different scenarios. The goal was to ensure that the system produces correct outputs for given inputs and handles errors effectively.

5.2. UNIT TESTING (Flask Backend Routing)

Unit testing focuses on testing individual components or modules of the system to ensure they function correctly in isolation. In this project, unit testing was performed on:

- Flask routes (login, registration, dashboard, test execution)
- Backend functions such as risk calculation
- Database operations (insert, update, fetch)

Each function was tested with valid and invalid inputs to ensure proper handling of data and error conditions. This helped in identifying bugs at an early stage and improving code reliability.

5.3. INTEGRATION TESTING (JavaScript Payloads to SQL Insertions)

Integration testing verifies the interaction between different modules of the system. It ensures that data flows correctly from frontend to backend and then to the database. In this system, integration testing was performed for:

- Frontend forms sending data to Flask backend
- Game inputs (clicks, responses) being processed and stored in the database
- Backend responses being displayed correctly on the frontend

For example, when a child completes a test, the frontend sends performance data to the backend, which processes and stores it in the database. The system was tested to ensure that this entire flow works without errors.

5.4. FUNCTIONAL TESTING (Web Speech API and Audio Validations)

Functional testing ensures that all features of the system work according to the specified requirements. The following functionalities were tested:

- **Game-based tests:**
  Verified whether reading, attention, and behavior tests function correctly
- **Web Speech API:**
  Checked whether audio instructions are generated properly and used effectively
- **User interface:**
  Ensured that buttons, forms, and navigation work correctly
- **Dashboard:**
  Verified whether charts and reports are displayed accurately.

This testing ensured that the system delivers the expected outputs for all user actions.

5.5. SECURITY TESTING (Bcrypt Password Hashing & Sessions)

Security testing ensures that the system is protected from unauthorized access and data breaches. The following security aspects were tested:

- **Password protection:**
  User passwords are securely handled using hashing techniques
- **Session management:**
  Ensures that users remain logged in securely and unauthorized access is prevented
- **Role-based access control:**
  Admin, Teacher, and Parent have restricted access based on their roles.

5.6. USER ACCEPTANCE TESTING (UAT)

User Acceptance Testing (UAT) is performed to evaluate whether the system meets user requirements and provides a good user experience. In this project, UAT focused on:

- Ease of use for teachers and parents
- Clarity of reports and dashboards
- Interaction of children with game-based tests
- Responsiveness of UI on different devices

Special attention was given to younger users (ages 1–5) to ensure that touch interactions, visuals, and instructions are easy to understand.

CHAPTER - 6

SYSTEM IMPLEMENTATION AND MAINTENANCE

Implementation is the phase where the designed system is developed and deployed into a working environment. It involves coding, integrating modules, configuring the environment, and ensuring that the system operates as expected. Maintenance refers to the continuous monitoring, updating, and improvement of the system after deployment.

The MindTrack: Dyslexia & ADHD Evaluator system is implemented as a web-based application using a lightweight and efficient technology stack. The system integrates frontend, backend, and database components to provide a seamless user experience.

6.1. SYSTEM IMPLEMENTATION PHASES

The system was developed in multiple phases to ensure proper planning, execution, and testing.

1. Requirement Analysis

In this phase, the problem of early detection of Dyslexia and ADHD was studied. User requirements for Admin, Teacher, Parent, and Child were identified. The need for gamified assessments and analytical dashboards was defined.

2. System Design

The architecture, modules, data flow, and database structure were designed. The system followed the MVC pattern and included modules such as authentication, game-based testing, dashboard, and training library.

3. Development Phase

The system was implemented using HTML, CSS, JavaScript for the frontend and Flask for the backend. Game-based tests were developed using HTML5 Canvas and Web Speech API. Database integration was done using SQL Alchemy.

4. Testing Phase

All modules were tested individually and as a complete system. Errors were identified and fixed to ensure smooth functionality.

5. Deployment Phase

The application was deployed in a local environment and can be accessed through a web browser. The system is designed to be easily deployable on cloud platforms if required.

6.2. ENVIRONMENT SETUP & DEPENDENCY INSTALLATION (requirements.txt)

The system requires a proper development environment for execution.

Software Setup:

- Install Python (latest version)
- Install Node.js (for frontend dependencies if required)
- Install a web browser (Chrome recommended)

Backend Setup:

- Navigate to backend folder
- Install required packages using:

```
pip install -r requirements.txt
```

- Run the Flask server:

```
python app.py
```

Frontend Setup:

- Open the frontend folder
- Run the development server (if using Vite/React):

```
npm install
npm run dev
```

The application will run locally and can be accessed through a browser using the provided URL.

6.3. ALGORITHM IMPLEMENTATION (RISK CALCULATION)

The system uses a rule-based algorithm to evaluate the child's performance and determine risk levels.

Key Parameters Used:

- Accuracy of responses
- Reaction time
- Commission errors (wrong actions)
- Omission errors (missed actions)

- Memory performance

Working of Algorithm:

1. Data is collected from game-based tests
2. Scores are calculated for reading, attention, and behavior
3. Scores are compared with predefined thresholds
4. Risk levels are assigned as:
    - Low Risk
    - Medium Risk
    - High Risk

This algorithm ensures consistent and unbiased evaluation of cognitive performance.

6.4. CROSS-BROWSER COMPATIBILITY & MAINTENANCE STRATEGY

The system is designed to work across different web browsers to ensure accessibility.

Cross-Browser Compatibility:

- Tested on Google Chrome, Edge, and Firefox
- Responsive design supports different screen sizes
- Touch support for mobile and tablet devices

Maintenance Strategy:

- Regular updates to fix bugs and improve performance
- Monitoring system usage and feedback
- Updating training modules and UI features
- Ensuring database integrity and security

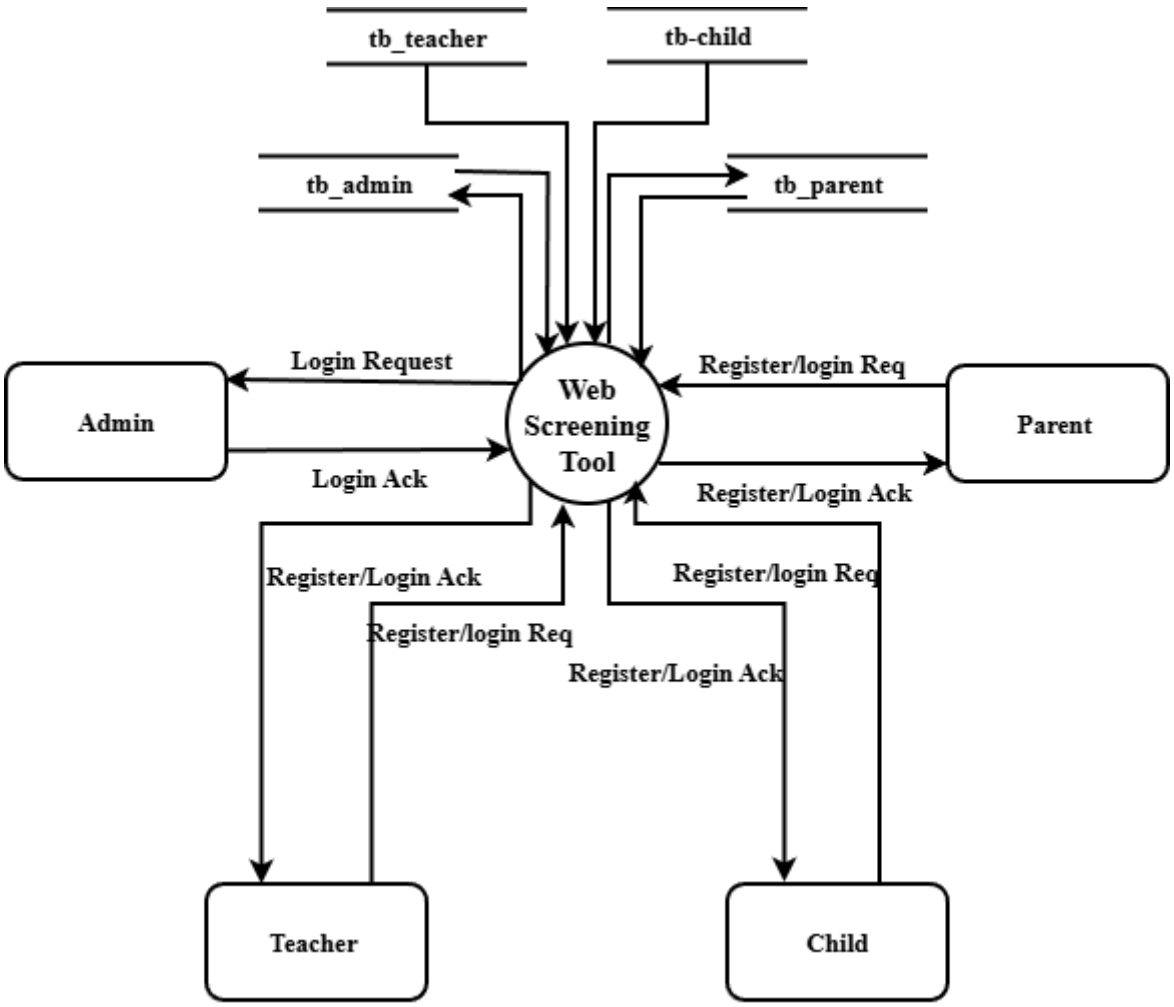
The system is built using a lightweight architecture, making it easy to maintain, update, and scale in the future.

CHAPTER - 7

APPENDIX

7.1. SYSTEM FLOW DIAGRAM

LEVEL 0



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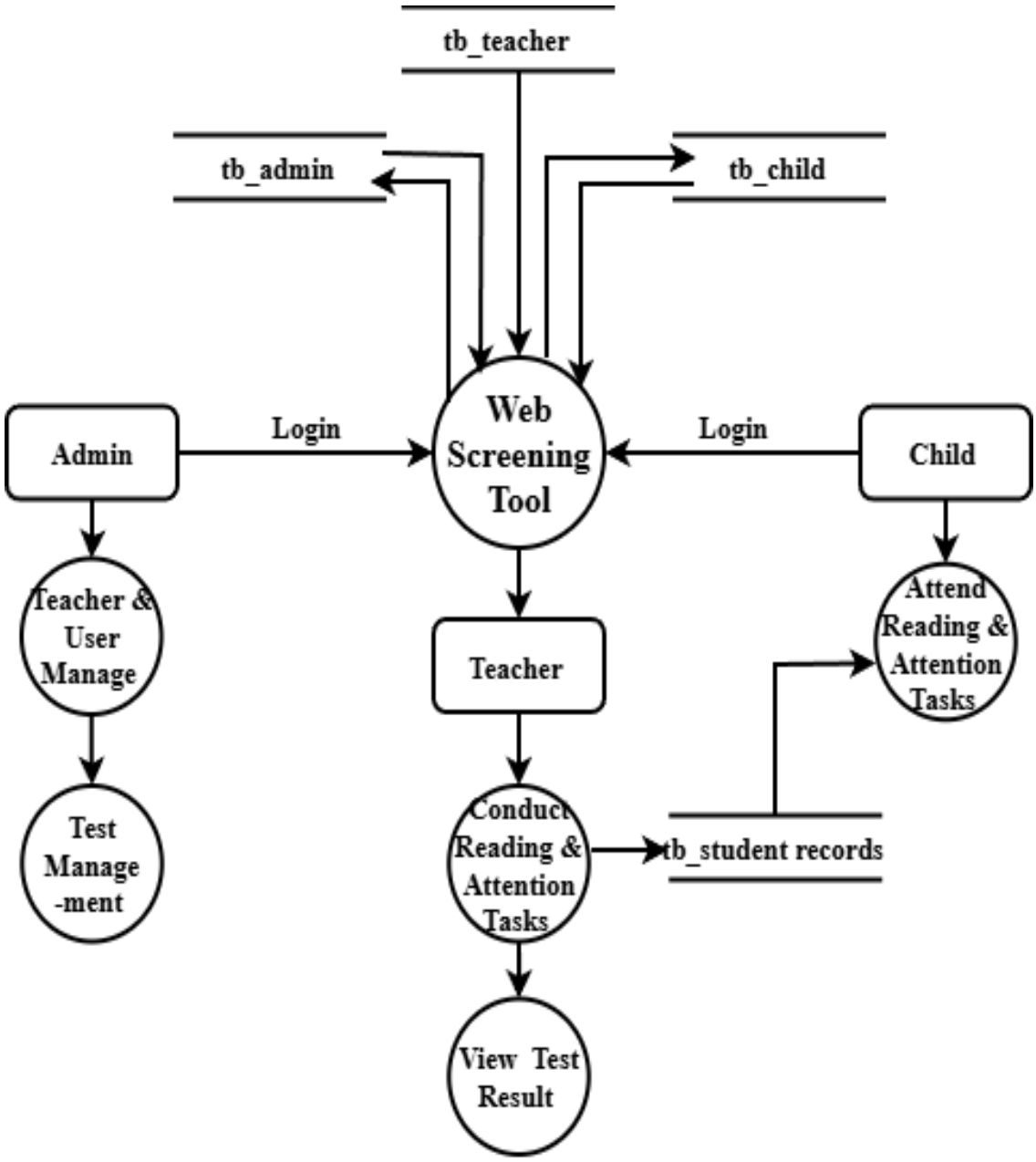
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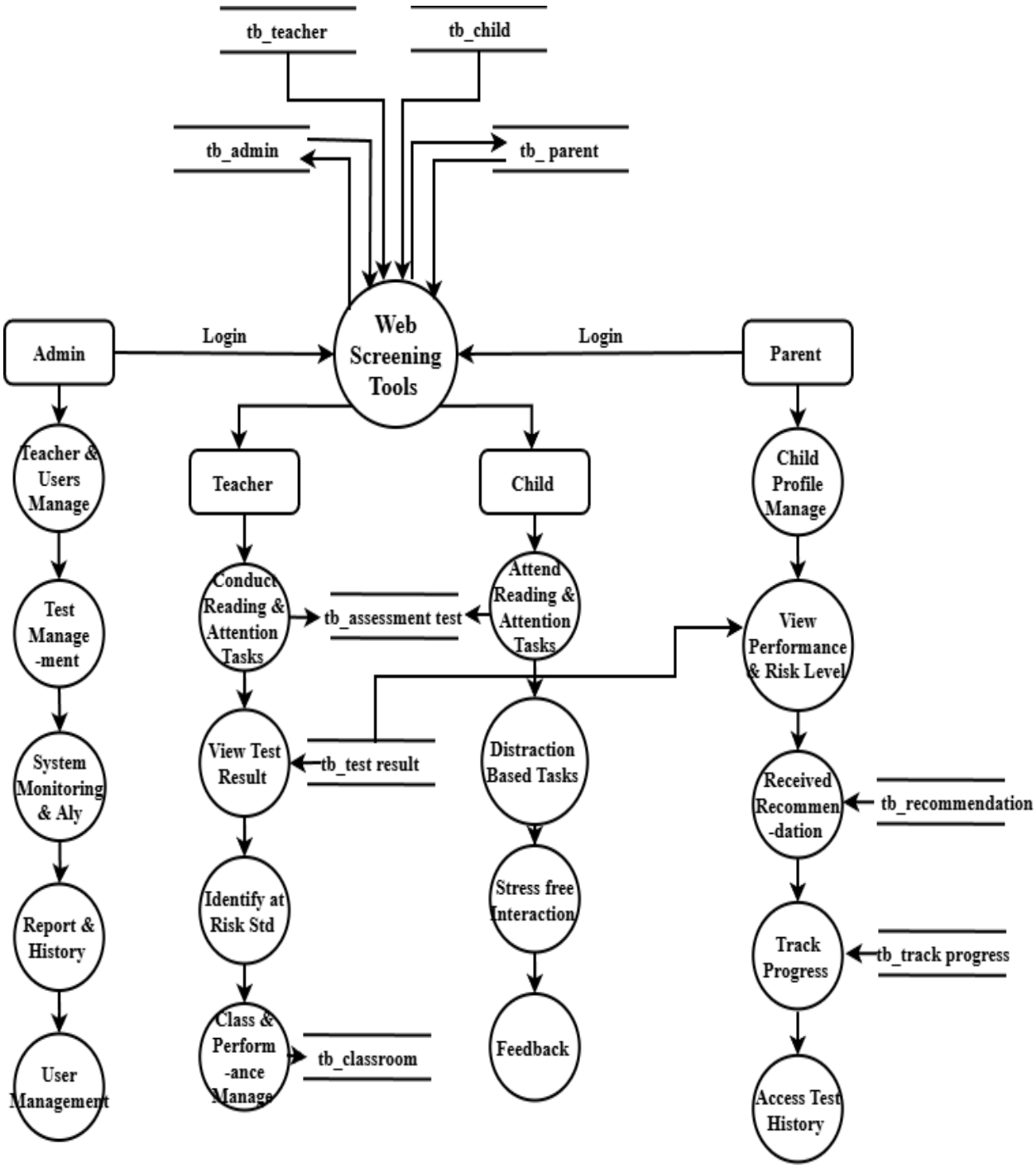
tb_admin
tb_child
Admin
Login
Web Screening Tool
Login
Child
Teacher & User Manage
Teacher
Attend Reading & Attention Tasks
Test Manage -ment
Conduct Reading & Attention Tasks
tb_student records
View Test Result

LEVEL 2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tb_child
tb_admin
tb_ parent
Admin
Login
Web Screening Tools
Login
Parent
Teacher & Users Manage
Teacher
Child
Child Profile Manage
Test Manage -ment
Conduct Reading & Attention Tasks
tb_assessment test
Attend Reading & Attention Tasks
View Performance & Risk Level
System Monitoring & Aly
View Test Result
tb_test result
Distraction Based Tasks
Received Recommen -dation
tb_recommendation
Report & History
Identify at Risk Std
Stress free Interaction
Track Progress
tb_track progress
User Management
Class & Perform -ance Manage
tb_classroom
Feedback
Access Test History

7.2. SAMPLE SCREENSHOTS

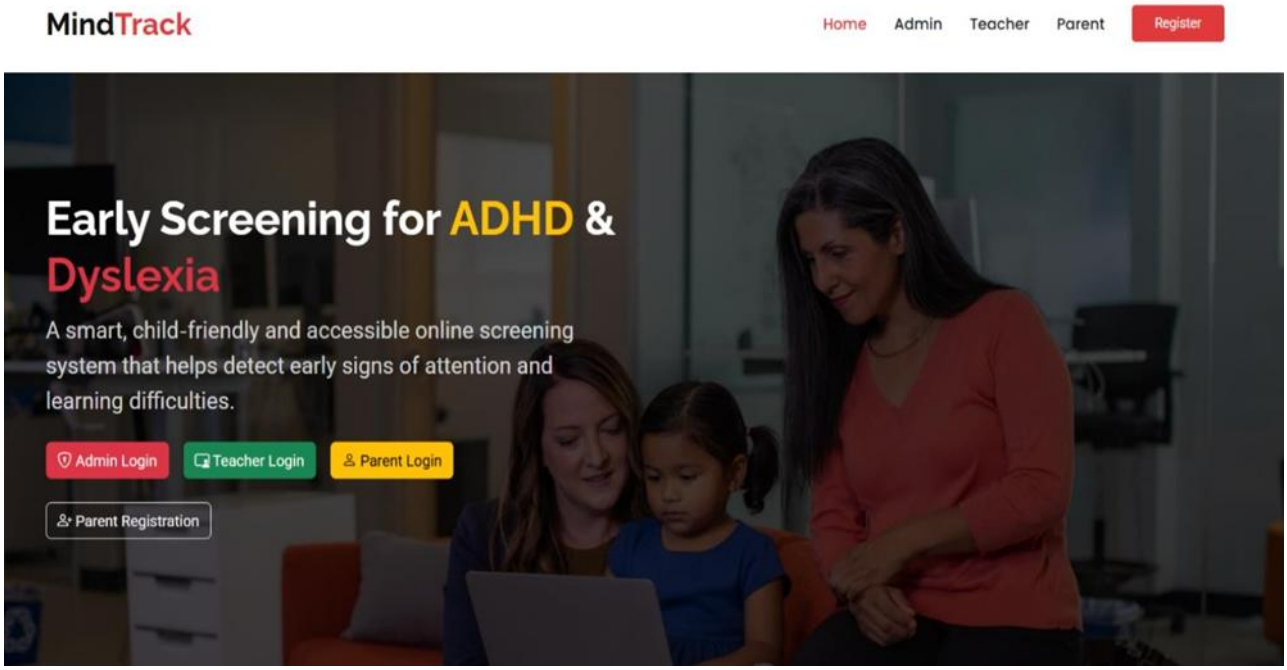

Figure 1: Home Page of MindTrack System

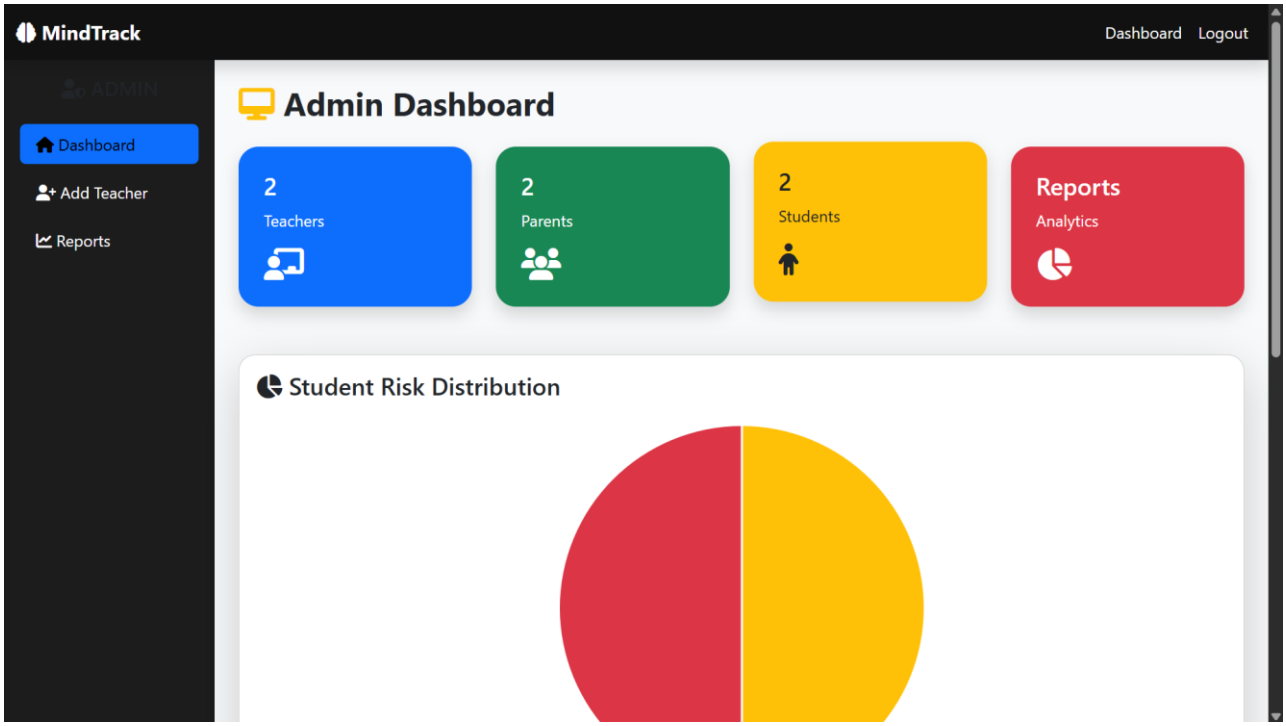

Figure 2: Admin Dashboard displaying analytics

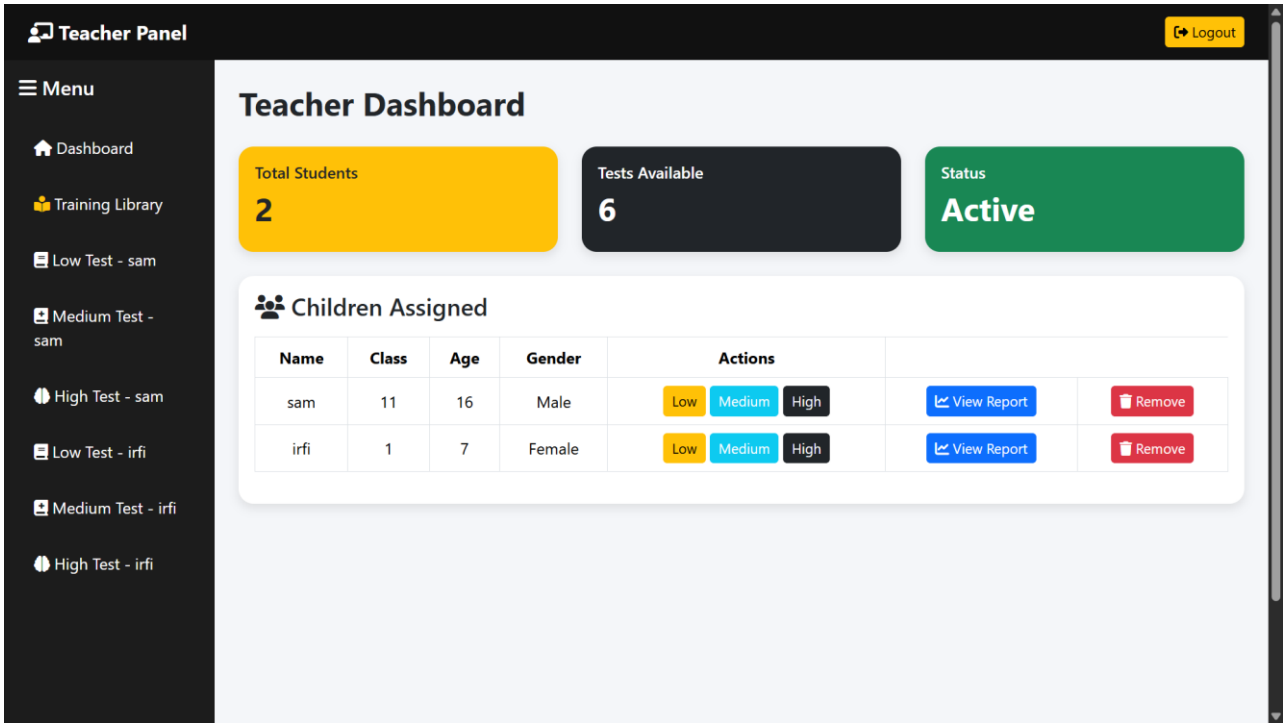

Figure 3: Teacher Dashboard for managing students

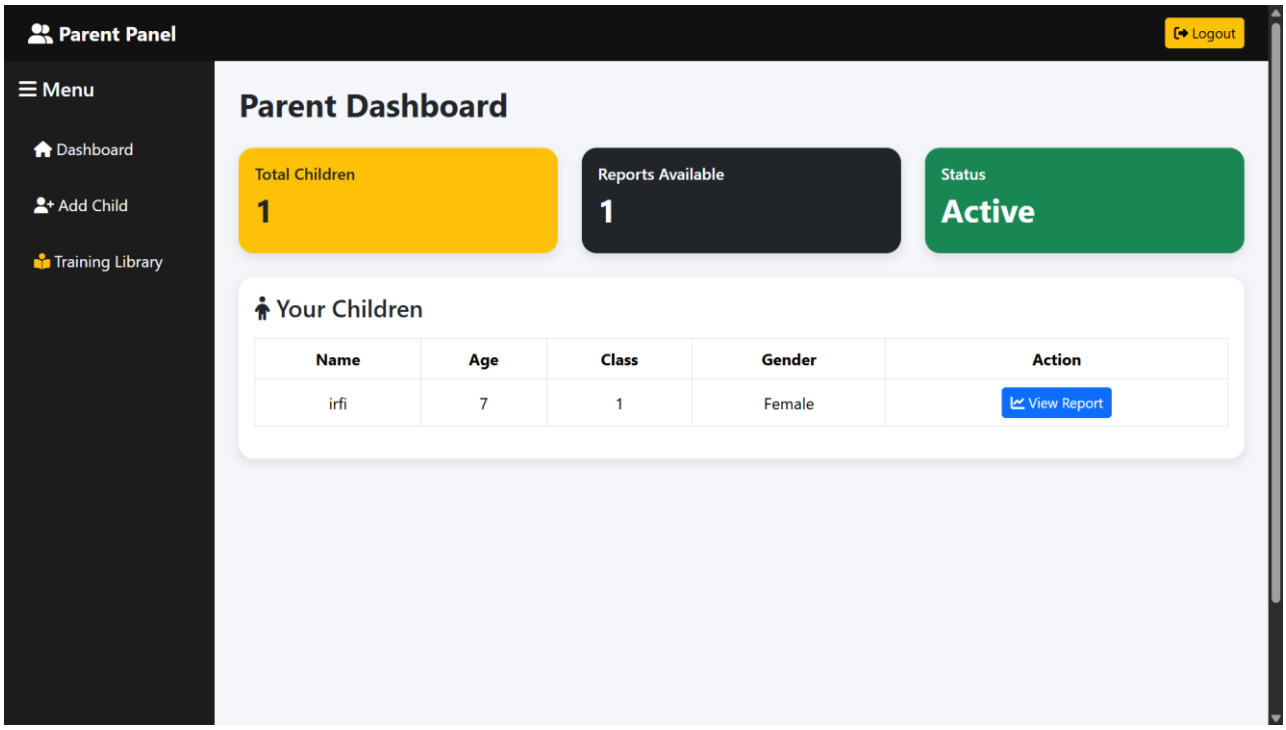

Figure 4: Parent Dashboard

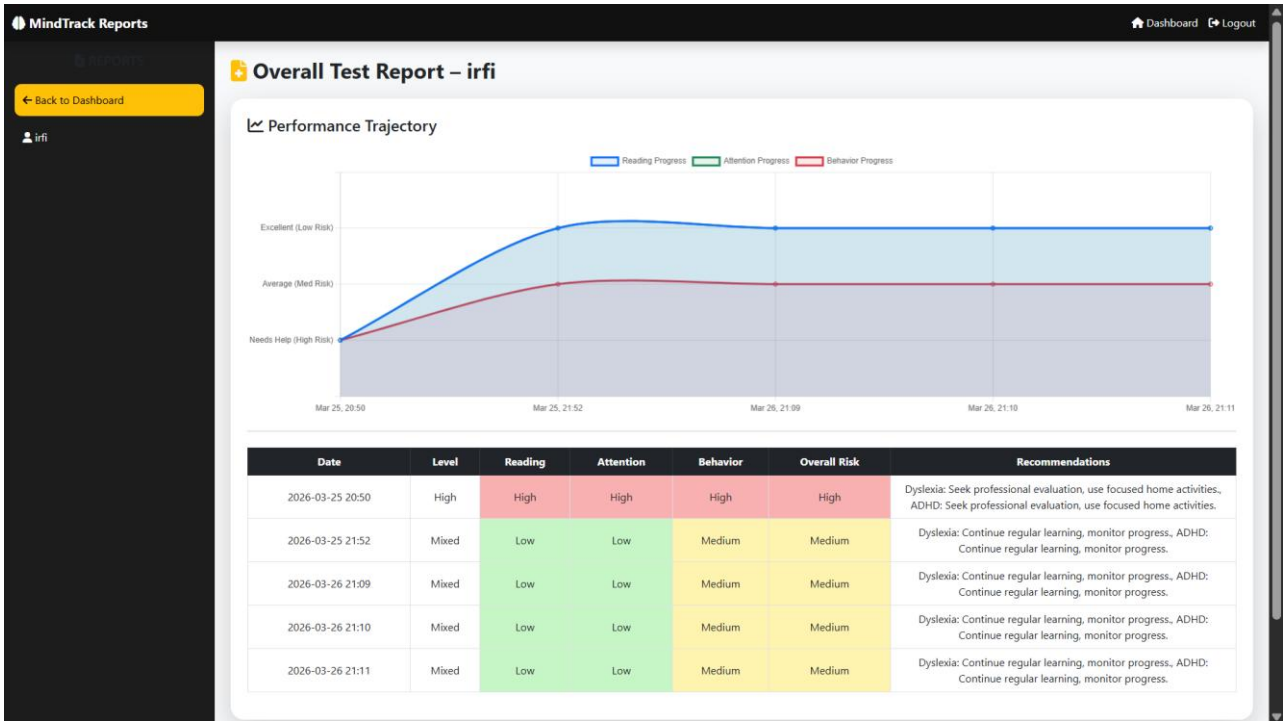

Figure 5: Performance Report Visualization

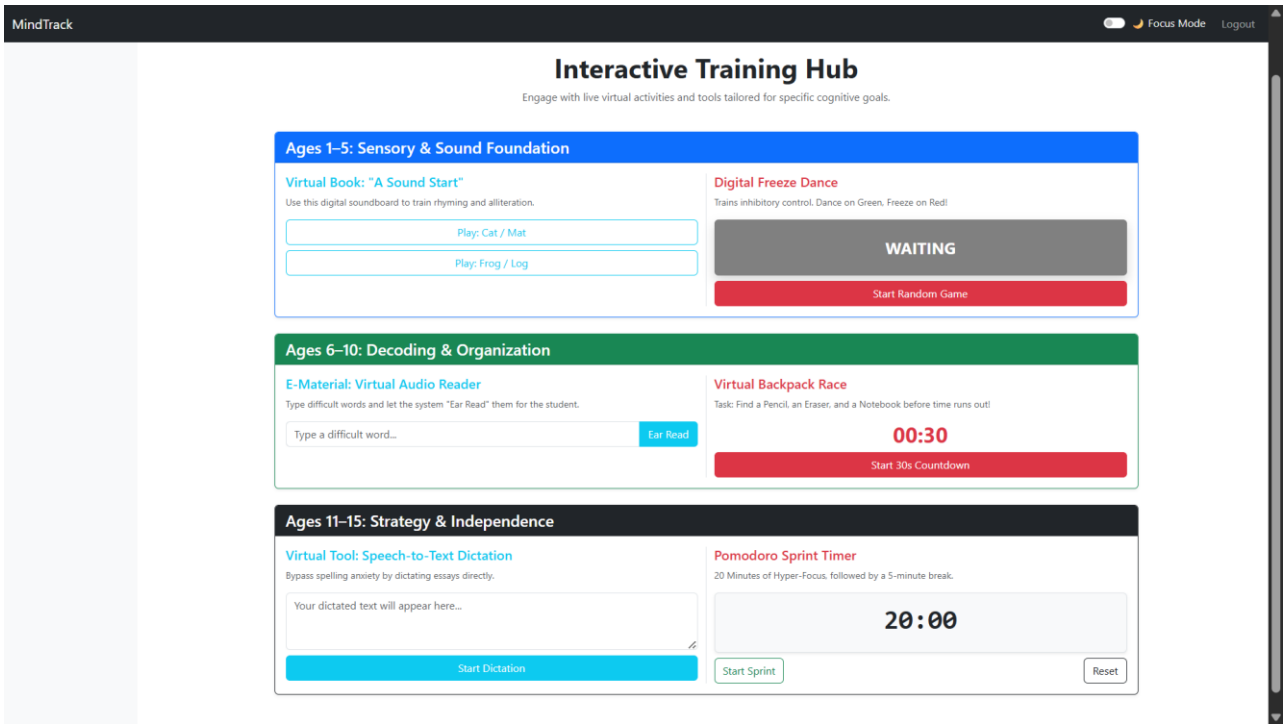

Figure 6: Training Module Interface

7.3. SAMPLE SOURCE CODE

PACKAGES

1. Backend Python Packages (Server-Side)

- **Flask**: The core lightweight web framework used to handle routing, server requests, and rendering HTML templates.
- **Flask-SQL Alchemy**: The Object Relational Mapper (ORM) used to securely interact with the database using Python objects instead of raw SQL queries.
- **Werkzeug.security**: A cryptographic library used to securely hash and verify user passwords (generate_password_hash, check_password_hash).
- **python-dotenv**: Used to securely load environment variables (like secret keys) from the .env file instead of hardcoding them into the source code.
- **datetime**: A native Python library used to capture exact timestamps for when physiological tests are completed.

2. Frontend Technologies & Libraries (Client-Side)

- **Chart.js**: A powerful JavaScript library imported via CDN, used to render the interactive, multi-axis Line Charts and Pie Charts on the Analytics Dashboards.
- **Canvas-Confetti**: A JavaScript library imported via CDN used to render the gamified "Dopamine Reward" particle system upon successful test submission.
- **Web Speech API (Native)**: A native browser API (window.speechSynthesis) used for Text-to-Speech (TTS) to audibly speak phonemes and instructions to younger children who cannot read.
- **HTML5 Canvas (Native)**: Used for precise, high-speed 2D rendering of the Continuous Performance Task (CPT) objects (like floating bubbles or falling shapes).
- **Bootstrap 5**: A CSS framework imported via CDN used to build the responsive navigation bar, Sidebar layouts, and Flash Message alert boxes.

CODING

1) BACKEND ROUTING & SECURE AUTHENTICATION (auth_service.py)

```
from werkzeug.security import generate_password_hash, check_password_hash


def register_parent(data):
    # Encrypt the password securely before persistent storage
```

```
    hashed_password = generate_password_hash(data['password'],
method='pbkdf2:sha256')

    new_user = Parent(
        name=data['name'],
        email=data['email'],
        mobile=data['mobile'],
        dob=datetime.strptime(data['dob'], "%Y-%m-%d"),
        username=data['username'],
        password=hashed_password
    )
    db.session.add(new_user)
    db.session.commit()
    return new_user
```

2) DYNAMIC 3x3 GAME ENGINE STATE GENERATOR (app.py)

```
def generate_game_data(age, level):
    # Determine the matrix cohort
    if age <= 5: age_group = '1-5'
    elif age <= 10: age_group = '6-10'
    else: age_group = '11-15'

    # Generate age-stratified environmental logic
    game_data = {
        'reading': {},
        'attention': {},
        'behavior': {}
    }

    if age_group == '1-5':
        # Prompt generation for 'Searching for Sounds' and 'Bubble Catch'
        phonemes = [('A', 'Apple'), ('B', 'Ball'), ('C', 'Cat')]
        target = random.choice(phonemes)
        game_data['reading']['target_letter'] = target[0]
        game_data['reading']['target_prompt'] = f"Which one starts with
{target[0]}?"
```

```
        game_data['attention']['target_color'] = random.choice(["RED", "BLUE",
"GREEN"])
        game_data['attention']['target_prompt'] = f"Pop the
{game_data['attention']['target_color']} ones!"

        game_data['behavior']['hidden_cup'] = random.randint(0, 2)

    return age_group, game_data
```

3) CANVAS-BASED COGNITIVE TESTING LOGIC (conduct_test.html)

```
// The Bubble Catch Javascript Engine (Age 1-5 Attention DOM)
startBubbleBtn.onclick = () => {
    bubblesActive = true;

    // Web Speech API triggers audio instructions
    const synth = window.speechSynthesis;
    const u = new SpeechSynthesisUtterance(gameData.attention.target_prompt);
    synth.speak(u);

    bubbleInterval = setInterval(spawnBubble, 1000);

    setTimeout(() => {
        clearInterval(bubbleInterval);
        bubblesActive = false;

        // Algorithmic Risk Scoring based on human error metrics
        if (impulseErrors > 2 || focusMisses > 2) attentionScore = 1;
        else if (impulseErrors > 0 || focusMisses > 0) attentionScore = 2;
        else attentionScore = 3;

        updateScoreUI("attentionScore", attentionScore);
    }, 15000);
};
```

4) ANALYTICS & VISUALIZATION ENGINE (parent_child_report.html)

```
// Destroy old overlapping charts to prevent canvas memory leaks
if (window.myLineChart) { window.myLineChart.destroy(); }

const ctx = document.getElementById('progressChart').getContext('2d');
window.myLineChart = new Chart(ctx, {
    type: 'line',
    data: {
        labels: testDates,
        datasets: [
            { label: 'Reading Score', data: readingScores, borderColor:
'#3498db', tension: 0.3 },
            { label: 'Attention Score', data: attentionScores, borderColor:
'#e74c3c', tension: 0.3 },
            { label: 'Behavior Score', data: behaviorScores, borderColor:
'#2ecc71', tension: 0.3 }
        ]
    },
    options: {
        responsive: true,
        maintainAspectRatio: false,
        plugins: {
            title: { display: true, text: 'Chronological Cognitive Trajectory'
}
        }
    }
});
```

CHAPTER - 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1. CONCLUSION

The MindTrack: Dyslexia & ADHD Evaluator system has been successfully designed and implemented as a web-based platform for early screening and monitoring of cognitive and learning disorders in children. The system effectively replaces traditional assessment methods with interactive, gamified evaluation techniques, making the process more engaging and suitable for children of different age groups.

By utilizing Continuous Performance Tasks (CPT) through game-based modules, the system captures real-time data such as response time, accuracy, and behavioral patterns. This data is processed using a rule-based algorithm to determine risk levels, which are presented in an easy-to-understand format using graphical dashboards. The inclusion of role-based access ensures that Admins, Teachers, and Parents can securely interact with the system. Additionally, the integration of a virtual training library enhances the system by providing tools for cognitive improvement, making it not only a detection system but also a support platform. Overall, the system is efficient, user-friendly, and capable of assisting in early identification and continuous tracking of Dyslexia and ADHD.

8.2. FUTURE ENHANCEMENTS

Although the current system provides a complete solution, it can be further improved by adding advanced features and expanding its capabilities.

- **Mobile Application Development:**
  The system can be extended into a mobile application using frameworks such as React Native or Flutter, enabling easy access on smartphones and tablets.
- **Machine Learning Integration:**
  Advanced machine learning models can be incorporated to improve prediction accuracy by analyzing large datasets and identifying complex behavioral patterns.

- **Cloud-Based Deployment:**
  Implementing cloud storage and deployment will enhance scalability, data security, and remote accessibility.
- **Advanced Analytics and Personalization:**
  The system can be enhanced with detailed analytics and personalized recommendations based on individual performance trends.
- **Multi-Language Support:**
  Adding support for multiple languages will improve accessibility for users from different regions.

CHAPTER - 9

BIBLIOGRAPHY AND WEB REFERENCES

BOOK REFERENCE

The following books, research materials, and online resources were referred to during the development of the MindTrack: Dyslexia & ADHD Evaluator system:

1. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, Pearson Education.
2. Kurose, J. F., & Ross, K. W., *Computer Networking: A Top-Down Approach*, Pearson.
3. Pressman, R. S., *Software Engineering: A Practitioner's Approach*, McGraw-Hill.

REFERENCES

- The following online resources and documentation were used for implementation and understanding technologies:
- Python Official Documentation – https://www.python.org/doc/
- Flask Documentation – https://flask.palletsprojects.com/
- SQL Alchemy Documentation – https://www.sqlalchemy.org/
- MDN Web Docs (HTML, CSS, JavaScript) – https://developer.mozilla.org/
- Chart.js Documentation – https://www.chartjs.org/docs/
- Web Speech API Documentation – https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API
- Bootstrap Documentation – https://getbootstrap.com/docs/
- W3Schools Web Development Tutorials – https://www.w3schools.com/
- Research articles on ADHD and Dyslexia screening methods (various academic sources)